

Soroush Vahidi
Department of Computer Science
Ying Wu College of Computing
New Jersey Institute of Technology
Newark, NJ 07102, USA

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Editor-in-Chief
Simulation Modelling Practice and Theory

Dear Editor-in-Chief,

I am pleased to submit the manuscript “A Trace-Driven Simulation Methodology for Testing LLM-Serving Scheduler Ranking Portability” for consideration as an original research article in *Simulation Modelling Practice and Theory*.

The manuscript studies a modelling and simulation question that arises when simulation is used to compare scheduling alternatives: whether the comparative ranking produced by a simulation study remains stable when the workload source, operating region, or evaluation metric changes. We address this question for large language model serving schedulers through LSSP, a preregistered, trace-driven deterministic discrete-event simulation study over three independently released workload sources, 120 workload windows, six calibrated operating regions, and a fixed policy panel.

The contribution is methodological rather than a proposal for a new scheduler. The paper formulates comparative ranking portability $R(W, M, L)$ as an object of simulation-based selection and comparison, uses a paired experimental design over matched workload windows, applies uncertainty-aware reversal detection with practical margins and confidence intervals, and estimates the sample complexity required for ranking recovery. The study also includes workload characterization, source separability analysis, and a selected physical vLLM/GPU validation.

The physical validation is reported as a model-fidelity result: the largest simulator-predicted selected reversal did not reproduce on physical execution, while the stable-control ordering qualitatively persisted. This finding helps define the domain of validity of the simulation results rather than overstating simulator accuracy.

The manuscript fits SMPT’s scope through its focus on trace-driven discrete-event simulation, experimental design for comparing alternatives, selection and comparison procedures, validation and verification, simulation of computer systems, and reproducible simulation artifacts for AI-serving infrastructure. Public artifacts are available through GitHub, Hugging Face, and Zenodo:

<https://github.com/SoroushVahidi/llm-serving-scheduler-robustness-benchmark>
<https://huggingface.co/datasets/SoroushVahidi/llm-serving-scheduler-portability>
<https://doi.org/10.5281/zenodo.22306798>

This manuscript is original, is not under consideration elsewhere, and I approve its submission as the sole author. Required declarations, including competing-interest, funding, data/code availability, CRediT contribution, and AI-assisted-writing statements, are included in the manuscript.

Sincerely,

Soroush Vahidi
Ph.D. Candidate in Computer Science
Department of Computer Science
Ying Wu College of Computing
New Jersey Institute of Technology
sv96@njit.edu